ECOQUEST: GAME CONCEPT FLOW CHART

System State Machine, Biome Progression & Modal Hierarchy

Project: EcoQuest (Unity 6 URP)
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1.0 BOOT & INTRO

Splash Screen & Logo
Init Audio & Settings Managers

2.0 MAIN MENU SCREEN (Assets/EcoQuest/Main Menu.unity)

- ▶ PLAY GAME
- ⚙ SETTINGS
- ✕ QUIT GAME

Menu Features & FX

- MenuAmbientFX (Fireflies)
- MenuParallax (Cursor Shift)
- MenuCritter (Wildlife)
- Audio Mixer BGM Route
- Async LoadingScreen.Load

⚙ SETTINGS MODAL (SettingsPanelUI.cs)

GRAPHICS

- Quality Presets:
  - URP Low
  - URP Medium
  - URP High (Def)
  - URP Ultra
- Resolution Select
- Fullscreen Mode
- VSync Toggle

AUDIO (MIXER)

- Master Volume
  Slider: -80 to 0 dB
- Music Volume
  Slider: -80 to 0 dB
- SFX Volume
  Slider: -80 to 0 dB
- Decibel Conversion

GAMEPLAY

- Mouse Sensitivity
  Range: 0.1x to 3.0x
- Invert Look Y
- WASD Key Layout
- Sprint Hold/Toggle
- Camera Bobbing

APPLY & SAVE (PlayerPrefs) | CLOSE / BACK

Accessible anytime from Main Menu or In-Game Pause Menu

Start Game

3.0 WORLD 1: FOREST SCENE (Tutorial, Baseline & 3-Bin Sorting)

Step 1: Movement Onboarding

- SequentialWASDTutorial prompts W, A, S, D
- Tab key prompt for Quest HUD

Step 2: Guardian Dialogue & Pre-Quiz

- Talk to Guardian (DialogueData.cs)
- Pre-Quiz Baseline Assessment (No Penalties)

Step 3: Kinesthetic Waste Sorting Quest (QuestManager.cs)

BIODEGRADABLE (Green)	RECYCLABLE (Blue)	NON-BIODEGRADABLE
Banana Peel, Apple, Leaves	Plastic Bottle, Can, Paper	Styrofoam, Broken Glass
Target: 3 Items	Target: 3 Items	Target: 3 Items

• Physics Grab & Hold Socket • Correct: Pile Visual Growth • Wrong: Alert Popup & Eject

Penalty: Every 3 Mistakes = -10 Gold Deducted (PlayerCurrency)

Step 4: Post-Quiz & Learning Gain

- Post-Test Evaluation (QuizManager)
- Scorecard: Learning Gain Δ Calculated
- Award: 100-150 Gold + 5 Gold/Correct
- **Sets: ForestRewardClaimed = 1**

Step 5: TravelGate Verification

- TravelGate.TryTravelToCity() checks:
  1. Tutorial Complete (WASD + Tab)
  2. Trash Quest Done (9/9 sorted)
  3. Post-Quiz Finished & Reward Claimed
- **Talk to Explorer → Load City Scene**

Async Scene Transition (LoadingScreenController.cs)

IN-GAME OVERLAYS & UI STACK

PAUSE MENU (PauseMenuController.cs - Esc)

- Freezes Time.timeScale = 0 & releases cursor
- Resume Game • Open Settings Modal
- How to Play / Controls Card
- Return to Main Menu (with Confirm Modal) • Quit

QUEST LOG (Tab)

- Toggles Quest Tab Pill
- Category progress bars
- Active objectives counter
- Minimap trash markers

INVENTORY (B)

- Backpack 12-slot grid
- Food: Apple, Sandwich
- Consumables / Potions
- Restores Hunger & Stamina

ECO SHOP MODAL (UI_EcoShop.cs)

- Opened via Handyman NPC in City Scene
- 8 Functional Items: Apple (15g), Sandwich (30g), Bento (55g), Energy Drinks (25g/50g), Swift/Winged Boots (25g/50g)
- 8 Locked Slots (Padlocks) • Coin counter pill bound to PlayerCurrency
- Elevates survival vitals HUD so buffs remain visible

SURVIVAL HUD (UI_SurvivalStatsHUD.cs)

- Top-left Hunger & Stamina bars • Timed Buff Badges (Speed & Stamina Regen)

4.0 WORLD 2: CITY SCENE (Urban Simulation, Commerce & Advanced Quests)

Plaza Spawn & Simulation

- Spawn at Central Plaza Fountain
- CityTrafficManager: Cars, Trains, Balloons
- 30+ Pedestrians via CityWalkGraph.cs

Handyman NPC & Eco Shop

- Stationed at Green Mobile Kiosk
- Opens Eco Shop Modal (UI_EcoShop)
- Purchase Food & Potions using Gold Coins

Robin: Trash Detective Quest

- Checklist of 5 District Trash Items
- Park, Main St, Market, Riverside
- 4 Waste Categories (Bio, Recyc, Non-Bio, Haz)
- Star rating based on time and accuracy

Guardian: Litterbug Chase Quest

- Guardian NPC directs player to find Litterbug
- Litterbug runs sidewalk routes dropping trash
- Corner Litterbug & persuade through dialogue
- Return to Guardian to claim round rewards

Step 5: City Bonus Quiz Evaluation (CityBonusQuizAdapter.cs)

• Dynamically generates 5 questions from trash items collected by player
• Pulls item funFact & category quizExplanation • Awards +10 Gold/correct + 20 Gold Bonus

Sets CityBonusTaken = 1 & CityBestStars in PlayerPrefs

⚠ ARCHITECTURAL NOTE: CITY EXIT / TRANSITION STATUS

No exit gate or completion portal is implemented in City Scene. Game currently dead-ends.
Planned World 3: Beach Scene (Beach.unity) exists only as an empty 4-GameObject stub.

5.0 WORLD 3: BEACH SCENE (PLANNED EXPANSION)

Current Asset Status

- Scene Asset: Assets/EcoQuest/Beach.unity (0.08 MB)
- Content: 4 GameObjects total (Camera, Light, Playerr)
- Build Settings: OMITTED (Not in EditorBuildSettings)

Planned Pedagogical Design (Roadmap)

- Theme: Coastal Cleanup & Marine Ecology
- Waste: Marine plastics, ghost nets, microplastics
- Mechanics: Tidal debris spawning, beach bins
- NPC: Coastal Ranger / Marine Biologist
- Educational Quiz: Ocean pollution & wildlife impact
- Victory: Overall Eco Champion Certification
- **Final Credits & Return to Main Menu**

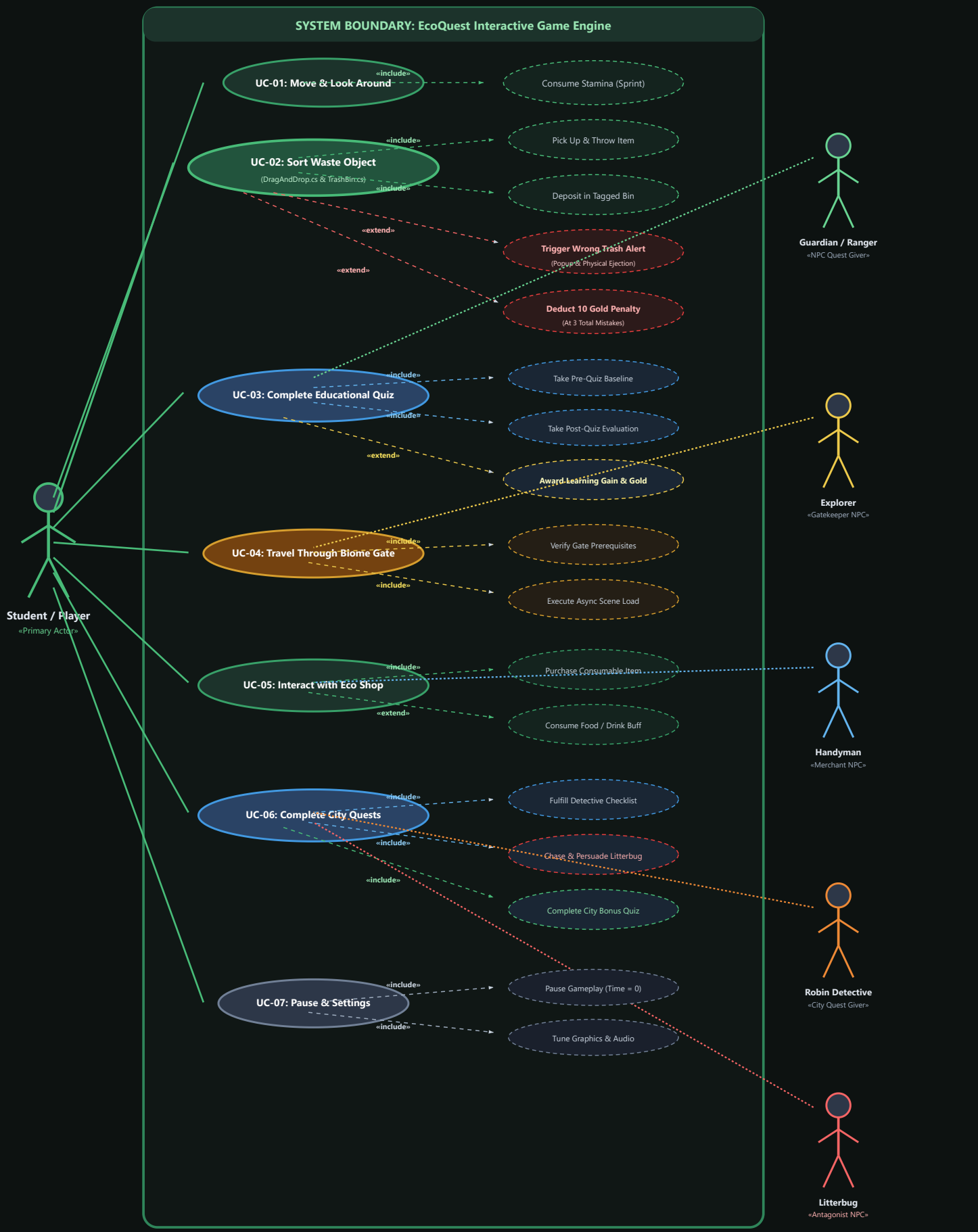
RECOMMENDED LEVEL TRANSITION FLOW

City Scene (Finish Quests) → Train/Harbor Travel Gate →
Beach Scene → Final Marine Post-Quiz → Victory Outro

ECOQUEST: UML USE CASE DIAGRAM

Actors, System Boundary, Core Mechanics & Extend/Include Relationships

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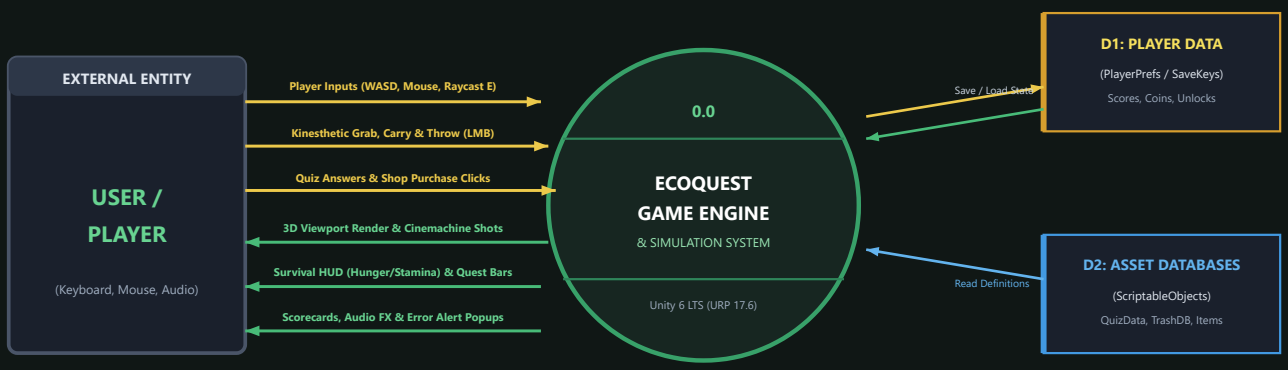


ECOQUEST: DATA FLOW DIAGRAMS (LEVEL 0 & LEVEL 1)

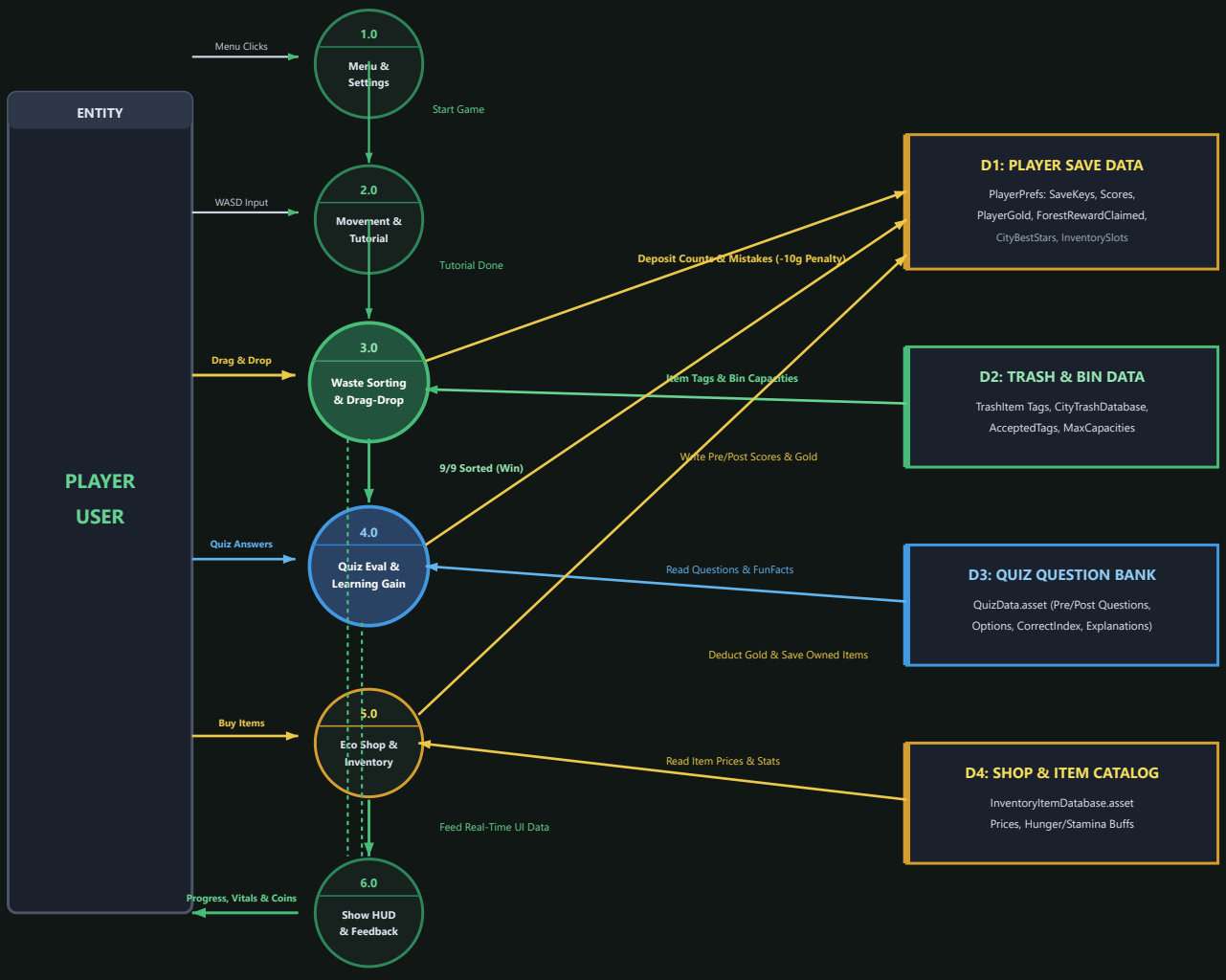
Context Boundary Diagram & Major Subsystem Process Flows

Project: EcoQuest (Unity 6 URP)
Author: Bryan Duarte (STI College Sta. Mesa)
Standard: Gane & Sarson / Yourdon DFD •

DATA FLOW DIAGRAM • LEVEL 0: CONTEXT DIAGRAM



DATA FLOW DIAGRAM • LEVEL 1: SYSTEM PROCESS DECOMPOSITION



ECOQUEST: DATA FLOW DIAGRAM (LEVEL 2 DECOMPOSITION)

Detailed Process Decomposition: Process 3.0 (Waste Sorting & Kinesthetic System)

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Scope: DragAndDrop & QuestManager Pipeline

